

Maternal Smoking Status and Infant Health

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Introduction

While abundant research concerning cigarette usage has garnered a negative public perception of smoking, targeted advertising of e-cigarettes to young adults has grown dramatically with the rise of social media usage¹. A 2015 survey of students in Los Angeles found that 30.7 percent of those who had used e-cigarettes started using combustible tobacco products during the first six months of the survey, compared to 8.1 percent who had never used e-cigarettes². Indeed, tobacco companies cater their marketing strategies to reach young audiences by leveraging sponsorships with popular brands such as Formula 1 and Spotify. As a result, the National Youth Tobacco Survey found that between 2022 and 2023, nearly 2.13 million students reported current use of e-cigarette usage, although teen tobacco usage had fallen prior to 2015³.

These statistics demonstrate the resurgence of a population of cigarette users, largely due to the tobacco industry's advertising to young adults by taking advantage of the population's tendency for brand loyalty and social media engagement. While the industry's primary audience has changed, the health concerns have not.

Our report illustrates one of the many health consequences of a lifetime of smoking—the impact of maternal smoking on infant weight. By using The Maternal Smoking and Infant Health data set collected from the larger 1960 to 1967 Child Health and Development Studies study, we examine whether mean infant weight is dependent on the number of years since the mother quit smoking. A regular consumer of nicotine will drastically reduce the supply of oxygen for the baby, due to nicotine's ability to constrict blood vessels in the body⁴. Thus, we anticipate that a greater, healthier infant weight will occur with greater number of years since maternal smoking cessation, or rather the absence of maternal smoking completely.

Upon examining whether infant weight is dependent on maternal smoking habits, its very impact on the mother's bodily health will be abundantly clear as well. As a new generation

¹(McKoy, 2024)

²(National Institutes of Health, 2015)

³(National Youth Tobacco Survey, 2023)

⁴(El-Ardat et al., 2014)

becomes the victim of the ever-evolving tobacco industry, it is imminent that we collect such research to cultivate a thorough understanding of smoking's spectrum of health detriments.

Exploratory Data Analysis

The Maternal Smoking and Infant Health data set is composed of information on babies and their parents, containing data on the parents and babies both from before birth and after birth. The data used in this lab is a small chunk of information extracted from *The Child Health and Development Studies*, a larger study focusing on pregnancies from 1960 to 1967. The women sampled were part of the Kaiser Foundation Health Plan in Oakland, California. All those enrolled in the same medicare program, Kaiser Plan, who delivered at any of the Kaiser hospitals in Northern California were included in the study. 5,000 families participated in the full study ranging from 1960 to 1957. (New York Times, 1995) *The Maternal Smoking and Infant Health* data that we use in this analysis spans from 1961 to 1962, and contains 1,236 participants. And upon excluding all N/A variables, we exclude 10 participants from the data set and are left with a filtered version of the set containing a total of 1,226 participants.

The filtered data set we use captures each baby's identification number, plurality, outcome (live birth or not), birth date, gestation (days), parity (total number of previous pregnancies including fetal deaths and stillbirths), and baby weight (oz). The parents' demographic information includes race, age (years), education, height (in), weight (pounds), marital status, and income of each mother and father. For the mothers, statistics are recorded concerning smoking status, the number of years since smoking cessation, and the number of cigarettes smoked per day for past and current smokers. These statistics are identifiable as 'smoke', 'time', and 'number' respectively in the data set.

To reiterate, the specific variables we examine in our test are time since mothers quit smoking (time) and mean baby weight (wt) after birth. This analysis will be done using an Analysis of Variance (ANOVA) test; a test that requires a categorical explanatory variable and a numerical response variable in order to compare means across two or more groups. Using the ANOVA test for our analysis allows us to compare mean baby weight across the range of times since maternal smoking cessation in order to examine the correlation between these two variables.

Thus, we select time since maternal smoking cessation (time) as our categorical variable and baby weight (wt) as our quantitative variable. We have selected these two variables under the name M_set, with 'time' as the explanatory variable and 'wt' as our response variable. While all baby weights were recorded, some maternal smoking statuses were not. To ensure all baby weights had an associated 'time' category, we filtered out the observations with an N/A value for 'time'. We would likely not gain much insight from keeping this data as our interpretations of the baby weights would vary greatly with the associated maternal smoking status. Otherwise, all other values from the baby weights and mother status were maintained in the M_set selection.

Results

Our research question aims to investigate the relationship between mean baby weight and the amount of time passed since maternal smoking cessation. If at least one mean baby weight across the time categories does differ, it is likely that we will have a statistically significant p-value. A statistically significant p-value is a probability value more extreme than our alpha value, the latter of which is a predetermined threshold for statistical significance. In our case, we choose an alpha value of 0.05 and will compare our p-value to determine the p-value's statistical significance. If we produce a p-value less than our alpha value 0.05, we can conclude that time baby weight is not dependent on the associated time since maternal smoking cessation. Before performing the test, we speculated that at least one of the mean baby weights across the time category would be different.

We are comparing the means of each baby weight (among the explanatory variable categories) in order to test the connection between these two variables. We will use the following hypothesis at an alpha level of 0.05.

- $H_0 : \mu_{ns} = \mu_{ss} = \mu_{dp} = \mu_1 = \mu_{1to2} = \mu_{2to3} = \mu_{3to4} = \mu_{5to9} = \mu_{10+} = \mu_q$
- $H_A : \text{At least one mean baby weight is different from another.}$

Key:

- **NS** = never smoked
- **SS** = still smokes
- **DP** = during current pregnancy
- **1** = within 1 year
- **1to2** = 1 to 2 years ago
- **2to3** = 2 to 3 years ago
- **3to4** = 3 to 4 years ago
- **5to9** = 5 to 9 years ago
- **10+** = 10+ years ago
- **q** = quit and don't know

Before conducting the ANOVA test, we must check whether our chosen variables meet the necessary conditions of independence, normality, and constant variance. If these conditions are not met, it will impact the validity of our ANOVA test and the resulting p-value, so we confirm that the conditions are met by plotting our variables in the following visualizations.

Table 1: The sample size n, mean, standard deviation, min, and max of mean baby weights among each category.

time	n	mean	sd	min	max
1 to 2 years ago	26	132.0769	14.75920	107	160

Table 1: The sample size n, mean, standard deviation, min, and max of mean baby weights among each category.

time	n	mean	sd	min	max
10+ years ago	6	113.0000	20.65914	84	143
2 to 3 years ago	12	117.7500	19.72135	65	140
3 to 4 years ago	9	124.1111	16.67666	103	146
5 to 9 years ago	15	120.4667	20.82535	75	148
during current preg	95	123.0842	17.80370	62	163
never smoked	544	122.7776	17.10966	55	176
still smokes	484	114.1095	18.09895	58	163
within 1 yr	30	124.8667	19.11802	90	170

Shown in Table 1, we can see the standard deviations listed for each explanatory group. From this we can confirm that the constant variance condition is met. The variability of the distributions of the response variable within each group have similar variance and the largest standard deviation (20.8 for the 5 to 9 group) is not more than double the smallest (14.8 for the 1 to 2 years group.)

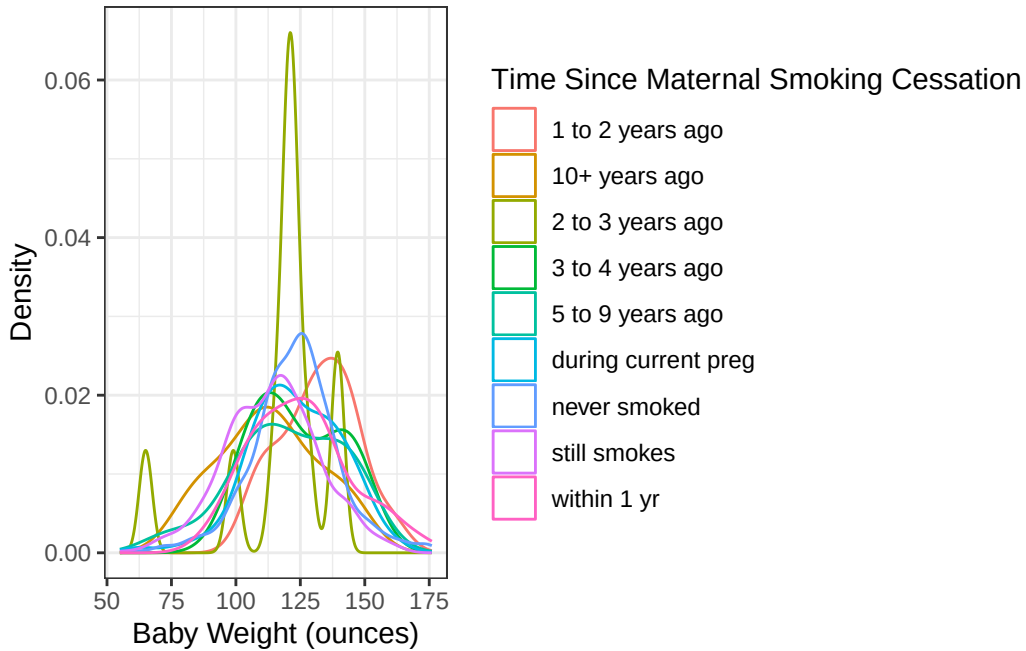


Figure 1: A density plot displaying which weights are most common among each group in the explanatory variable.

In Figure 1, we see the plots of each baby weight and time since quitting combo. Looking at

this we can claim that the normality condition is met because the distributions within each group are approximately normal. The 3 to 4 year distribution is the only plot that appear does not appear to be completely normal. It is too multi-modal so we will keep this in mind. The abnormality is likely due to the small sample size (9) that we can see in Figure 1. Although the sample size is small, it still meets conditions so we will just keep the plot in mind when testing and proceed with caution.

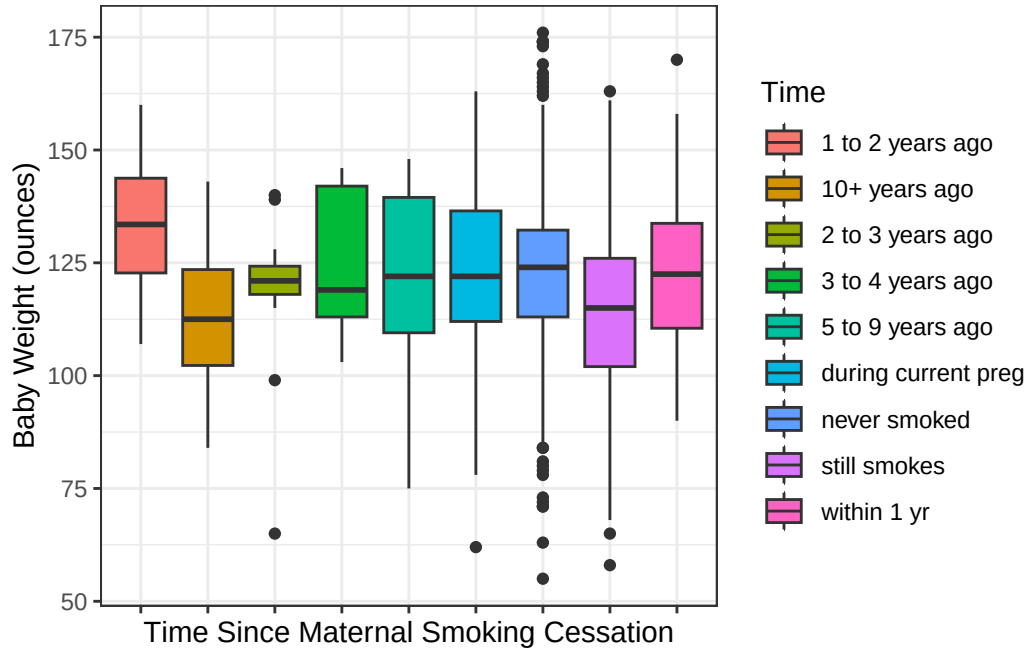


Figure 2: This figure visualizes the differences in baby weight between groups in the ‘time’ variable using boxplots.

Additionally, in Figure 2 we can see that some box plots of the categorical variable such as 2 to 3 years ago and never smoked have outliers. Because the sample size seen in Figure 1 for never smoked is so large (544) we can relax this condition. For graphs containing smaller sample sizes such as 2 to 3 years ago (12) we will keep the outliers in mind as we proceed with caution.

Finally, we can assume that the independence condition is met because within groups the sampled observations are independent of each other and between groups the groups are independent of each other as well. No mother is being counted for two groups (ex. never smoked and still smokes) and no baby weight is being counted twice.

Moving forward with the ANOVA test, we obtained were given a table containing mean squared error (MSE) and mean squared variation due to groups (MSG) which we used to calculate an F-statistic of 9.516. This value indicates the ratio of explained variance (the differences in mean baby weights due to time since maternal smoking cessation) to unexplained variance

(other factors influencing baby weights). The F statistic helps us understand how much of the variability in baby weights can be attributed to the time since the mother quit smoking.

$$F = \frac{MSG}{MSE} \approx \frac{2969}{312} \approx 9.516$$

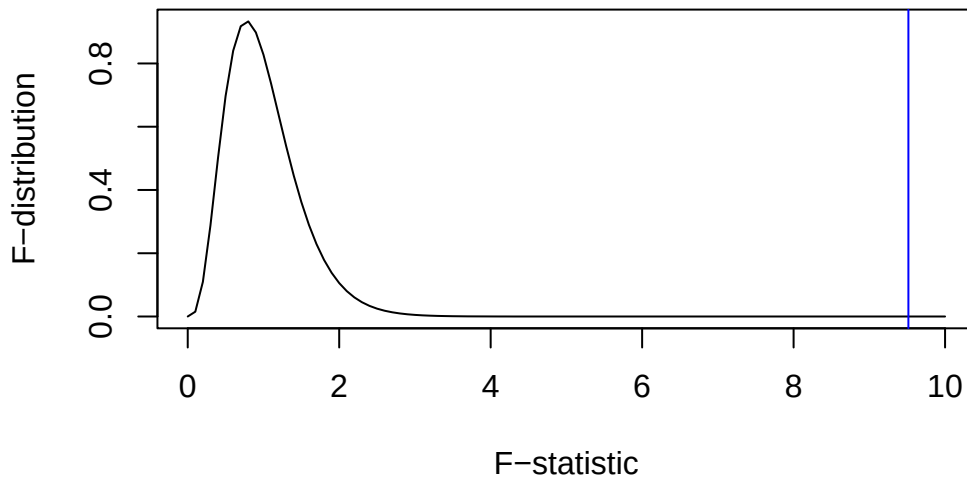


Figure 3: This F-distribution curve shows where our F-statistic lies in comparison to if the data set demonstrated that the wt variable was not dependent on the time variable.

```
# A tibble: 2 x 6
  term      df  sumsq meansq statistic  p.value
<chr>   <dbl>  <dbl>  <dbl>   <dbl>   <dbl>
1 time         8 26600.  3325.    10.7 1.41e-14
2 Residuals 1212 377725.   312.    NA    NA
```

The p-value is determined by measuring how likely our F statistic is to occur given a null distribution of the comparison of variability within and between groups. Our p-value associated with our F statistic in the table was approximately 0.000. This means that our data has a statistically significant difference in at least one mean baby weight. Because our alpha (the threshold for significance) was 0.05, any number below this indicates that the observed differences in baby weights among the different groups of time since smoking cessation are unlikely to have just occurred by chance. It appears that there is a significant difference in at

least one mean baby weight across groups of time since maternal smoking cessation, meaning that there is a dependent relationship between smoking and baby weight.

To see which two means had the largest difference, we performed an ANOVA Post Hoc test. This test produced a table displaying the adjusted p-value for each variable combination in the data. Our dataset shows mainly p-values of 1.000 (not significant) but a few statistically significant p-values as well (below 0.05). The smallest p-value, approximately 0.000, was calculated between the categories “still smokes” and “never smoked”. This tells us that these two categories have the largest difference in mean baby weights and are likely the main variable responsible for our statistically significant result. Although the following categories have greater p-values than the p-value between “still smokes” and “never smoked” categories, we observe that the categories “still smokes” and “during current pregnancy”, and “still smokes” and “1 to 2 years ago” also have statistically significant p-values.

Discussion

Upon examining the Maternal Smoking and Infant Health data set, we were interested in finding whether the time since the mother had quit smoking impacted an infant’s weight at birth. We performed an Analysis of Variance (ANOVA) test, and found that the true means of the infant weights could be equivalent across varying times passed since the mother quit smoking.

After rejecting our null hypothesis, our post-hoc test determined that the “never smoked” category and the “still smokes” category had the largest difference in sample means, with the pair having the smallest p-value in the post-hoc test.

Table 2: The differences in sample size (n) among each category of time since smoking cessation.

time	sample_size
1 to 2 years ago	26
10+ years ago	6
2 to 3 years ago	12
3 to 4 years ago	9
5 to 9 years ago	15
during current preg	95
never smoked	544
still smokes	484
within 1 yr	30

While these results were not explicit indicators of ineffective data, we found that the sampling technique was not conducive to producing the most reliable results. Having greater sample sizes

in each category would have produced less likelihood for error, allowing us to be more confident in our resulting statistics. While we do find that time since maternal smoking cessation impacts baby weight, we did not include a consideration for the variability in maternal age. As age can factor into the overall health of the mother and thus the capability to birth a healthier infant, the exclusion of the maternal age variable in our study is particularly important to note.

The implications of this study offer tangible real-world applications that show importance to public health. While it is widely acknowledged that smoking during pregnancy poses certain risks to both the mother and the child, during the 1960s, the era in which this data was collected, it was not. Links between cigarette smoking and cancer were just beginning to emerge in the 1950s, making a study like this extremely relevant to the times⁵. A study like ours demonstrating a statistically significant correlation between smoking status and baby weight would have provided invaluable support to important speculations being made at the time as well as inspiring further investigation into a pressing public health issue.

In modern times, its relevancy still stands as more information on the effects of smoking are being investigated, as social media and a new generation fuels a resurgence of smoking culture. With this data set specifically, crucial insights can be gained about the effect of smoking on unborn children. Using an ANOVA test allowed us to see that there is in fact a correlation between time since maternal smoking cessation and baby weight. In the future, this data set could be conducted once again, but for those who engage in “dual use” of tobacco products, such as e-cigarettes or cigars⁶. Not only did informational studies used from this data serve as a historic milestone in understanding the effects of smoking, but it also acts as a catalyst for continued inquiry on the specific effects of various forms of smoking on newborns.

Appendix

To refer back to Table 2, ideally, our sample sizes across explanatory category groups are large, and produce less Standard Error. As the Central Limit Theorem states, the greater the sample size, the more normally distributed the data will be. Thus, smaller sample sizes are less conducive to having a normal distribution, as a larger Standard Error value will result in a wider null distribution. While our test statistic may remain the same regardless of changes in sample sizes among groups, the associated p-value will change as the null distribution widens or condenses.

⁵(Sasco, 2004)

⁶(CDC, 2023)

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